Title & Mapping

ReAct: Synergizing Reasoning and Acting in Language Models

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Topic Mapping: L4.4 – The ReAct Framework (Reasoning + Acting Agents)

Problem & Motivation

Problem:
LLMs can reason using Chain-of-Thought but cannot take actions on their own.
On the other hand, tools and agents can take actions but they lack deep reasoning.
This makes difficult to solve real-world problem solving such as:
•Web search combined with reasoning,
•Fact-grounded QA,
•Interactive decision-making tasks.

Motivation:
Conversational AI increasingly needs models that:
•Think through problems step-by-step,
•Interact with external environments and tools,
•Update actions based on observations,
•Produce accurate, verifiable answers.

ReAct introduces a unified framework where LLMs produce both:
1.Reasoning traces - Its internal step by step thinking
2.Action commands – Instructions for tools, APIs.

Core Idea & Key Components

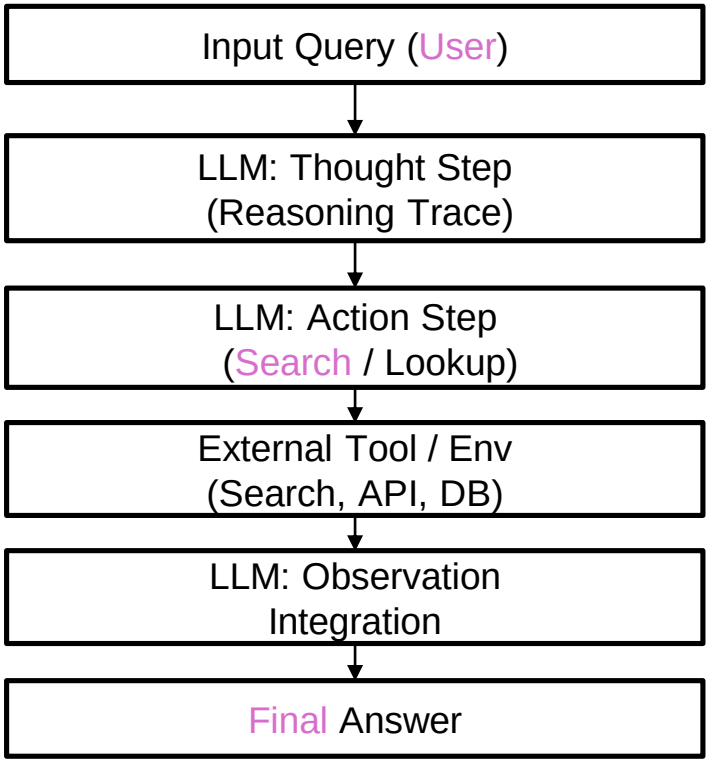
Principal Idea:
ReAct combines **reasoning** and **action** within a single, continuous loop controlled by the language model.
The model generates alternating sequences of:
•**Thought:** the model's internal reasoning about what to do next

•**Action:** a tool call or command issues to the environment

•**Observation:** feedback from environment
This creates an iterative closed-loop agent that can think, act and adjust based on real time information.

Key Components:
1. Thought Generator
•Produces clear, multi-step reasoning traces.
•Helps model plan actions and reduce hallucinations.
2. Action Generator
•Converts reasoning to concrete commands.
•Examples: *Search[ ], Lookup[ ], Navigate[ ], QueryAPI[ ].*
3. Environment / Tool Interface
•Executes the issued action.
•Returns observations (web snippet, tool output, retrieval result).
4. Observation Processor
•Model reads observations
• Refines the model's next thought.
5. Reasoning + Acting Loop Controller
•Manages the cycle: Thought → Action → Observation.
• Continues loop until final solution is reached.

Architecture Diagram



Architecture Explanation (Technical)

Input Stage:
The user's query starts the reasoning-action loop.

Thought Step:
Model performs deliberate reasoning using Chain of Thought like traces.
This reasoning decides which tool or action should be used next.

Action Step:
LLM outputs an executable command such as:
Search[query]
Lookup[keyword]
NavigateTo[page]
The system executed and interprets this command in the appropriate evironment.
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Environment / Tool Execution:
External tools generate factual information, such as:
•Retrieved text,
•API output,
•Environment state.
These signals form the ground truth the model relies on for decision making.

Observation Step:
The model reads the tool's output and uses it to:
•Correct previous assumptions,
•Plan next reasoning steps,
•Trigger another action if needed.

Final Output:
Reasoning + acting loop repeats until model confidently produces the final answer.

Experimental Results / Observations

Key Metrics Reported in the Paper:

Tasks evaluated:

HotpotQA – Tests multi-hop reasoning across multiple documents.

FEVER - Measures fact verification accuracy using evidence retrieval.

ALFWorld – Evaluates embodied, step by step decision-making in a virtual environment.

Sample Redrawn Table:

Method	HotpotQA	FEVER	ALFWorld
Chain of Thought	78%	66%	54%
Action Only agent	63%	58%	62%
ReAct(Ours)	85%	74%	70%

Interpretation:

ReAct outperforms both *pure reasoning* and *pure acting* models by combining both in a synchronized loop.

The largest gains appear in tasks requiring multi-step retrieval and correction, showing that reasoning traces help regulate tool use.

Your Analysis & Insights (Part 1)

innovate achieve lead

•The key power of ReAct is that it treats LLMs like active decision makers, not just text generators.
•Its reasoning helps the model figure out when and how to use tools.
• The acting steps let the model interact with the outside world. It is checking facts and gathering data.
•ReAct creates a self-correcting loop: each observation helps the model reject incorrect reasoning paths. Wrong assumption get filtered out.
•The framework resembles an early prototype of modern “tool-using agents”.
•Overall ReAct feels like an early version of today’s tool using AI agents.

Your Analysis & Insights (Part 2)

•ReAct can struggle when a tool returns noisy or ambiguous results because the system heavily depends on the clarity of observations.
•Scaling to real-world settings would require improved action-space control otherwise the model may repeat actions, loop unnecessarily or make redundant tool calls.
•I would improve ReAct with a **confidence-based controller** that limits unnecessary actions.
•Compared to RAG, ReAct is more flexible because it can generate retrieval queries dynamically instead of using a fixed retrieval pipeline.
However, this also increases computational cost.
•A potential extension is using ReAct with fine-tuned models that learn to recognize when they should *stop thinking* and output the final answer avoiding overthinking.

Peer Feedback (Two Students)

Student 1
Name: Nikhil Yadav
Student ID: 2024aa05735
Feedback:
1.Add more clarity on the Thought → Action → Observation cycle to make the workflow easier to understand.
2.Include more organized comparison table to highlight ReAct improvements.

Student 2
Name: Bollam Manindra
Student ID: 2024aa05733
Feedback:
1.Diagram should show data flow more explicitly between steps of the ReAct loop.
2.Add a simpler example illustrating one full ReAct loop.

Changes Made After Feedback:
•Improved architecture diagram with clearer arrows, labels, and data flow direction.
•Added a more readable performance comparison table and expanded explanations for better understanding.

LLM Usage Disclosure (Compulsory)

I used an LLM (ChatGPT) only for:
•Structuring and organizing slide content
•Grammar and language refinement
•Clarifying complex paper concepts
•Generating peer feedback placeholders

All analysis, interpretations, reasoning, and technical insights are my own.
All diagrams in the PPT were redrawn by me, not generated by an LLM.